```
mysql> -- CSA05 DBMS ASSIGNMENT CO2
Query OK, 0 rows affected (0.002 sec)

mysql> -- SWIFTSPAN PARCELS
Query OK, 0 rows affected (0.002 sec)

mysql> -- COMPLETE MYSQL IMPLEMENTATION
Query OK, 0 rows affected (0.002 sec)

mysql> -- ================================================
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> DROP DATABASE IF EXISTS SwiftSpan;
Query OK, 0 rows affected, 1 warning (0.177 sec)

mysql> CREATE DATABASE SwiftSpan;
Query OK, 1 row affected (0.109 sec)

mysql> USE SwiftSpan;
Database changed
mysql>
mysql> -- ================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- 1. NORMALIZED DESIGN (BCNF)
Query OK, 0 rows affected (0.002 sec)

mysql> -- ================================================
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> -- SENDER TABLE
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> CREATE TABLE SENDER (
    ->     SenderID INT PRIMARY KEY,
    ->     SenderName VARCHAR(100) NOT NULL,
    ->     RouteGroup VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.356 sec)
```

```
mysql>
mysql> CREATE TABLE SERVICE (
    ->     ServiceID INT PRIMARY KEY,
    ->     ServiceCode VARCHAR(30) UNIQUE NOT NULL,
    ->     ServiceName VARCHAR(100) NOT NULL,
    ->     VolumeWeight DECIMAL(10,2) NOT NULL
    -> );
Query OK, 0 rows affected (0.376 sec)

mysql>
mysql> -- COORDINATOR TABLE
Query OK, 0 rows affected (0.003 sec)

mysql>
mysql> CREATE TABLE COORDINATOR (
    ->     CoordinatorID INT PRIMARY KEY,
    ->     CoordinatorName VARCHAR(100) NOT NULL,
    ->     Hub VARCHAR(50) NOT NULL
    -> );
Query OK, 0 rows affected (0.254 sec)

mysql>
mysql> -- ASSIGNMENT TABLE
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> CREATE TABLE ASSIGNMENT (
    ->     SenderID INT,
    ->     ServiceID INT,
    ->     CoordinatorID INT NOT NULL,
    ->     PRIMARY KEY (SenderID, ServiceID),
    ->     FOREIGN KEY (SenderID) REFERENCES SENDER(SenderID),
    ->     FOREIGN KEY (ServiceID) REFERENCES SERVICE(ServiceID),
    ->     FOREIGN KEY (CoordinatorID) REFERENCES COORDINATOR(CoordinatorID)
    -> );
Query OK, 0 rows affected (0.444 sec)

mysql>
mysql> -- COMPLETION POLICY TABLE
Query OK, 0 rows affected (0.005 sec)
```

```
mysql>
mysql> -- OBSERVATION TABLE
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> CREATE TABLE OBSERVATION (
    ->     SenderID INT,
    ->     ServiceID INT,
    ->     DispatchWave INT,
    ->     Completion INT NOT NULL,
    ->     PRIMARY KEY (SenderID, ServiceID, DispatchWave),
    ->     FOREIGN KEY (SenderID, ServiceID)
    ->         REFERENCES ASSIGNMENT(SenderID, ServiceID),
    ->     FOREIGN KEY (Completion)
    ->         REFERENCES COMPLETION_POLICY(Completion),
    ->     CHECK (Completion BETWEEN 0 AND 100)
    -> );
Query OK, 0 rows affected (0.701 sec)

mysql>
mysql> -- ================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- 2. INSERT SAMPLE DATA
Query OK, 0 rows affected (0.001 sec)

mysql> -- ================================================
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> INSERT INTO SENDER VALUES
    -> (1,'Arun Traders','North'),
    -> (2,'Bala Stores','South'),
    -> (3,'Chennai Exports','East'),
    -> (4,'Delta Logistics','West'),
    -> (5,'Evergreen Shop','North');
Query OK, 5 rows affected (0.056 sec)
Records: 5  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO SERVICE VALUES
```

```
mysql> INSERT INTO ASSIGNMENT VALUES
    -> (1,1,1),(1,2,1),(1,3,2),(1,4,2),(1,5,3),
    -> (2,1,1),(2,2,2),(2,3,2),(2,4,3),(2,5,3),
    -> (3,1,2),(3,2,2),(3,3,1),(3,4,1),(3,5,3),
    -> (4,1,1),(4,2,3),(4,3,3),(4,4,2),(4,5,1),
    -> (5,1,2),(5,2,1),(5,3,2),(5,4,3),(5,5,1);
Query OK, 25 rows affected (0.068 sec)
Records: 25  Duplicates: 0  Warnings: 0

mysql>
mysql> -- Completion percentage policy
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> INSERT INTO COMPLETION_POLICY VALUES
    -> (0,'Low'),(10,'Low'),(20,'Low'),
    -> (30,'Low'),(38,'Low'),(39,'Medium'),
    -> (40,'Medium'),(50,'Medium'),
    -> (60,'Medium'),(70,'High'),
    -> (80,'High'),(90,'High'),(100,'High');
Query OK, 13 rows affected (0.079 sec)
Records: 13  Duplicates: 0  Warnings: 0

mysql>
mysql> INSERT INTO ROUTE_POLICY VALUES
    -> ('Low',1.50),
    -> ('Medium',1.20),
    -> ('High',1.00);
Query OK, 3 rows affected (0.249 sec)
Records: 3  Duplicates: 0  Warnings: 0

mysql>
mysql> -- OBSERVATIONS
Query OK, 0 rows affected (0.004 sec)

mysql>
mysql> INSERT INTO OBSERVATION VALUES
    -> (1,1,1,80),(1,1,2,90),(1,1,3,70),(1,1,4,85),
    -> (1,2,1,60),(1,2,2,70),(1,2,3,80),(1,2,4,90),
    -> (1,3,1,50),(1,3,2,60),(1,3,3,70),(1,3,4,80),
    -> (1,4,1,40),(1,4,2,50),(1,4,3,60),(1,4,4,70),
    -> (1,5,1,30),(1,5,2,40),(1,5,3,50),(1,5,4,60),
    ->
```

```
mysql>
mysql> SELECT
    ->     S.SenderID,
    ->     S.SenderName,
    ->     V.ServiceName,
    ->     O.DispatchWave,
    ->     O.Completion,
    ->     V.VolumeWeight,
    ->     P.ServiceBand,
    ->     R.RouteFactor,
    ->     (O.Completion * V.VolumeWeight * R.RouteFactor)
    ->         AS Weighted_Adjusted_Index
    -> FROM OBSERVATION O
    -> JOIN SENDER S ON O.SenderID = S.SenderID
    -> JOIN SERVICE V ON O.ServiceID = V.ServiceID
    -> JOIN COMPLETION_POLICY P ON O.Completion = P.Completion
    -> JOIN ROUTE_POLICY R ON P.ServiceBand = R.ServiceBand
    -> WHERE O.SenderID = 1
    -> AND O.DispatchWave = 4;
Empty set (0.008 sec)

mysql>
mysql>
mysql> -- QUERY 2:
Query OK, 0 rows affected (0.002 sec)

mysql> -- SENDERS MEETING THRESHOLD 39
Query OK, 0 rows affected (0.001 sec)

mysql> -- FOR EVERY REQUIRED SERVICE
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT S.SenderID, S.SenderName
    -> FROM SENDER S
    -> WHERE NOT EXISTS (
    ->     SELECT 1
    ->     FROM ASSIGNMENT A
    ->     WHERE A.SenderID = S.SenderID
    ->     AND NOT EXISTS (
    ->         SELECT 1
```

```
    -> FROM OBSERVATION O
    -> JOIN SENDER S ON O.SenderID = S.SenderID
    -> JOIN SERVICE V ON O.ServiceID = V.ServiceID
    -> WHERE O.DispatchWave = 4
    -> AND NOT EXISTS (
    ->     SELECT 1
    ->     FROM OBSERVATION X
    ->     WHERE X.SenderID = O.SenderID
    ->     AND X.ServiceID = O.ServiceID
    ->     AND X.DispatchWave = 4
    ->     AND X.Completion < 39
    -> );
Empty set (0.010 sec)

mysql>
mysql>
mysql> -- QUERY 4:
Query OK, 0 rows affected (0.007 sec)

mysql> -- COORDINATOR MEANS BELOW HUB MEAN
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT
    ->     C.CoordinatorID,
    ->     C.CoordinatorName,
    ->     AVG(O.Completion) AS Coordinator_Mean
    -> FROM COORDINATOR C
    -> JOIN ASSIGNMENT A
    -> ON C.CoordinatorID = A.CoordinatorID
    -> JOIN OBSERVATION O
    -> ON A.SenderID = O.SenderID
    -> AND A.ServiceID = O.ServiceID
    -> GROUP BY C.CoordinatorID, C.CoordinatorName
    -> HAVING AVG(O.Completion) < (
    ->     SELECT AVG(O2.Completion)
    ->     FROM OBSERVATION O2
    ->     JOIN ASSIGNMENT A2
    ->     ON O2.SenderID = A2.SenderID
    ->     AND O2.ServiceID = A2.ServiceID
    ->     JOIN COORDINATOR C2
    ->     ON A2.CoordinatorID = C2.CoordinatorID
    ->     WHERE C2.Hub = C.Hub
```

```
mysql>
mysql> SELECT * FROM SENDER;
+----------+-----------------+-------------+
| SenderID | SenderName      | RouteGroup |
+----------+-----------------+-------------+
|        1 | Arun Traders    | North      |
|        2 | Bala Stores     | South      |
|        3 | Chennai Exports | East       |
|        4 | Delta Logistics | West       |
|        5 | Evergreen Shop  | North      |
+----------+-----------------+-------------+
5 rows in set (0.002 sec)

mysql> SELECT * FROM SERVICE;
+-----------+-------------+------------------+--------------+
| ServiceID | ServiceCode | ServiceName      | VolumeWeight |
+-----------+-------------+------------------+--------------+
|         1 | EXP01       | Express Delivery |         2.50 |
|         2 | STD02       | Standard Delivery |         3.00 |
|         3 | ECO03       | Economy Delivery |         4.50 |
|         4 | DOC04       | Document Delivery |         1.00 |
|         5 | FRG05       | Fragile Delivery |         5.00 |
+-----------+-------------+------------------+--------------+
5 rows in set (0.002 sec)

mysql> SELECT * FROM COORDINATOR;
+---------------+-----------------+------------+
| CoordinatorID | CoordinatorName | Hub        |
+---------------+-----------------+------------+
|             1 | Kumar           | Chennai    |
|             2 | Ravi            | Coimbatore |
|             3 | Mani            | Madurai    |
+---------------+-----------------+------------+
3 rows in set (0.002 sec)

mysql> SELECT * FROM ASSIGNMENT;
+----------+-----------+---------------+
| SenderID | ServiceID | CoordinatorID |
+----------+-----------+---------------+
|        1 |         1 |             1 |
|        1 |         2 |             1 |
|        2 |         1 |             1 |
|        3 |         3 |             1 |
```

```
mysql>
mysql> SELECT COUNT(*) AS Total_Observations
    -> FROM OBSERVATION;
+--------------------+
| Total_Observations |
+--------------------+
|                  0 |
+--------------------+
1 row in set (0.005 sec)

mysql>
mysql> -- SUM
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT SUM(Completion) AS Total_Completion
    -> FROM OBSERVATION;
+------------------+
| Total_Completion |
+------------------+
|             NULL |
+------------------+
1 row in set (0.002 sec)

mysql>
mysql> -- AVERAGE
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT AVG(Completion) AS Average_Completion
    -> FROM OBSERVATION;
+--------------------+
| Average_Completion |
+--------------------+
|               NULL |
+--------------------+
1 row in set (0.002 sec)

mysql>
mysql> -- MAXIMUM
Query OK, 0 rows affected (0.001 sec)

mysql>
```

```
mysql>
mysql> -- HAVING
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> SELECT SenderID, AVG(Completion) AS Average_Completion
    -> FROM OBSERVATION
    -> GROUP BY SenderID
    -> HAVING AVG(Completion) > 50;
Empty set (0.002 sec)

mysql>
mysql> -- ORDER BY
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SELECT * FROM OBSERVATION
    -> ORDER BY Completion DESC;
Empty set (0.004 sec)

mysql>
mysql>
mysql> -- ================================================
Query OK, 0 rows affected (0.001 sec)

mysql> -- 5. CREATE INDEXES
Query OK, 0 rows affected (0.001 sec)

mysql> -- ================================================
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> CREATE INDEX idx_observation_wave
    -> ON OBSERVATION(DispatchWave);
Query OK, 0 rows affected (0.197 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql> CREATE INDEX idx_observation_completion
    -> ON OBSERVATION(Completion);
Query OK, 0 rows affected (0.251 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql>
mysql> CREATE VIEW OBSERVATION_DETAILS AS
    -> SELECT
    ->     O.SenderID,
    ->     S.SenderName,
    ->     O.ServiceID,
    ->     V.ServiceName,
    ->     O.DispatchWave,
    ->     O.Completion
    -> FROM OBSERVATION O
    -> JOIN SENDER S ON O.SenderID = S.SenderID
    -> JOIN SERVICE V ON O.ServiceID = V.ServiceID;
Query OK, 0 rows affected (0.228 sec)

mysql>
mysql> -- DISPLAY VIEWS
Query OK, 0 rows affected (0.005 sec)

mysql>
mysql> SELECT * FROM SENDER_DETAILS;
+----------+----------------+------------+
| SenderID | SenderName     | RouteGroup |
+----------+----------------+------------+
|        1 | Arun Traders   | North      |
|        2 | Bala Stores    | South      |
|        3 | Chennai Exports | East      |
|        4 | Delta Logistics | West      |
|        5 | Evergreen Shop | North      |
+----------+----------------+------------+
5 rows in set (0.021 sec)

mysql> SELECT * FROM OBSERVATION_DETAILS;
Empty set (0.015 sec)

mysql>
mysql>
mysql> -- ================================================
Query OK, 0 rows affected (0.003 sec)

mysql> -- 7. UPDATE AND DELETE
Query OK, 0 rows affected (0.004 sec)

mysql> -- ================================================
```

```
ysql>
ysql> DELETE FROM OBSERVATION
    -> WHERE SenderID = 5
    -> AND ServiceID = 5
    -> AND DispatchWave = 4;
uery OK, 0 rows affected (0.008 sec)

ysql>
ysql>
ysql> -- ================================================
uery OK, 0 rows affected (0.003 sec)

ysql> -- 8. TRANSACTION TEST
uery OK, 0 rows affected (0.003 sec)

ysql> -- ================================================
uery OK, 0 rows affected (0.002 sec)

ysql>
ysql> START TRANSACTION;
uery OK, 0 rows affected (0.003 sec)

ysql>
ysql> UPDATE SERVICE
    -> SET ServiceName = 'Express Courier'
    -> WHERE ServiceID = 1;
uery OK, 1 row affected (0.006 sec)
ows matched: 1  Changed: 1  Warnings: 0

ysql>
ysql> ROLLBACK;
uery OK, 0 rows affected (0.032 sec)

ysql>
ysql>
ysql> -- ================================================
uery OK, 0 rows affected (0.002 sec)

ysql> -- 9. ROLLED-BACK ANOMALY TEST
uery OK, 0 rows affected (0.002 sec)
```

```
mysql>
mysql> CREATE TABLE TRACKING_CARD (
    ->     SenderID INT,
    ->     ServiceID INT,
    ->     DispatchWave INT,
    ->     SenderName VARCHAR(100),
    ->     ServiceName VARCHAR(100),
    ->     CoordinatorName VARCHAR(100),
    ->     Hub VARCHAR(50),
    ->     Completion INT,
    ->     ServiceBand VARCHAR(30),
    ->     RouteFactor DECIMAL(6,2),
    ->     PRIMARY KEY (SenderID, ServiceID, DispatchWave)
    -> );
Query OK, 0 rows affected (0.282 sec)

mysql>
mysql> -- POPULATE READ CARD
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> INSERT INTO TRACKING_CARD
    -> SELECT
    ->     O.SenderID,
    ->     O.ServiceID,
    ->     O.DispatchWave,
    ->     S.SenderName,
    ->     V.ServiceName,
    ->     C.CoordinatorName,
    ->     C.Hub,
    ->     O.Completion,
    ->     P.ServiceBand,
    ->     R.RouteFactor
    -> FROM OBSERVATION O
    -> JOIN SENDER S ON O.SenderID = S.SenderID
    -> JOIN SERVICE V ON O.ServiceID = V.ServiceID
    -> JOIN ASSIGNMENT A
    -> ON O.SenderID = A.SenderID
    -> AND O.ServiceID = A.ServiceID
    -> JOIN COORDINATOR C
    -> ON A.CoordinatorID = C.CoordinatorID
```

```
mysql> -- 11. CHECK DATABASE SIZE AND ROW COUNTS
Query OK, 0 rows affected (0.001 sec)

mysql> -- ================================================
Query OK, 0 rows affected (0.002 sec)

mysql>
mysql> SELECT COUNT(*) AS Sender_Count FROM SENDER;
+--------------+
| Sender_Count |
+--------------+
|            5 |
+--------------+
1 row in set (0.004 sec)

mysql> SELECT COUNT(*) AS Service_Count FROM SERVICE;
+---------------+
| Service_Count |
+---------------+
|             5 |
+---------------+
1 row in set (0.003 sec)

mysql> SELECT COUNT(*) AS Observation_Count FROM OBSERVATION;
+-------------------+
| Observation_Count |
+-------------------+
|                 0 |
+-------------------+
1 row in set (0.002 sec)

mysql> SELECT COUNT(*) AS Tracking_Card_Count FROM TRACKING_CARD;
+--------------------+
| Tracking_Card_Count |
+--------------------+
|                  0 |
+--------------------+
1 row in set (0.002 sec)
```

```
+---------------------+------------+-------------+--------------+
| TABLE_NAME          | TABLE_ROWS | DATA_LENGTH | INDEX_LENGTH |
+---------------------+------------+-------------+--------------+
| assignment          |         25 |       16384 |        32768 |
| completion_policy   |         13 |       16384 |            0 |
| coordinator         |          3 |       16384 |            0 |
| observation         |          3 |       16384 |        16384 |
| observation_details |       NULL |        NULL |         NULL |
| route_policy        |          3 |       16384 |            0 |
| sender              |          5 |       16384 |            0 |
| sender_details      |       NULL |        NULL |         NULL |
| service             |          5 |       16384 |        16384 |
| tracking_card       |          0 |       16384 |            0 |
+---------------------+------------+-------------+--------------+
10 rows in set (0.198 sec)

mysql>
mysql>
mysql> -- ================================================
Query OK, 0 rows affected (0.002 sec)

mysql> -- 12. FINAL DISPLAY
Query OK, 0 rows affected (0.001 sec)

mysql> -- ================================================
Query OK, 0 rows affected (0.001 sec)

mysql>
mysql> SHOW TABLES;
+---------------------+
| Tables_in_swiftspan |
+---------------------+
| assignment          |
| completion_policy   |
| coordinator         |
| observation         |
| observation_details |
| route_policy        |
| sender              |
| sender_details      |
| service             |
| tracking_card       |
+---------------------+
```